OFFICIAL COURSE PROPOSAL / PRACTICAL AI SYSTEMS

AI AUTOMATION WITH n8n

From zero foundations to practical, production-ready business automation



36	108	12	3 × 3h
SESSIONS	TOTAL HOURS	WEEKS	WEEKLY FORMAT

Build workflows • Integrate APIs • Add AI • Deploy responsibly

A practical path from manual work to intelligent systems

A project-led program focused on building useful automations and delivering them responsibly.

BUSINESS-FIRST

Identify repetitive work, map processes, and select valuable automation opportunities.

WORKFLOW ENGINEERING

Build maintainable workflows with triggers, logic, data transformation, and integrations.

AI WHERE IT ADDS VALUE

Use LLMs, structured outputs, retrieval, and controlled tools for language-heavy tasks.

PRODUCTION MINDSET

Test failures, protect credentials, monitor runs, and document handover.

Outcome: learners finish with portfolio projects and a deployable capstone—not just copied templates.

Course at a glance

The schedule follows the requested delivery format.

12 WEEKS

3 sessions every week

36 SESSIONS

Each session is 3 hours

108 HOURS

Total guided learning time

6 STAGES

Foundations to professional delivery

DELIVERY FORMAT

- Live explanation + guided build
- Hands-on lab in every session
- Weekly assignment or checkpoint
- Project work throughout

COURSE DESIGN PRINCIPLES

- Start at zero; no automation background required
- Learn concepts before adding complexity
- Use real APIs and business examples
- Practical production skills without extreme-scale topics

Who is this course designed for?

Zero-barrier entry, with a clear progression toward independent project delivery.

UNIVERSITY STUDENTS

Interested in AI, software, data, and practical digital skills.

ASPIRING AUTOMATION BUILDERS

Want to connect apps and remove repetitive tasks.

FREELANCERS & OPERATORS

Want to build and maintain automations for teams or small businesses.

PREREQUISITES

- Basic computer and browser use
- Comfortable following step-by-step instructions
- No prior n8n or AI-agent experience required

HELPFUL BUT NOT REQUIRED

- Basic spreadsheets and business processes
- Some Python / JavaScript familiarity
- Willingness to test, debug, and learn by building

What learners will be able to do

Measurable outcomes aligned with practical workflow delivery.

01 Design

Map a manual process into a trigger → logic → action workflow.

02 Build

Create workflows with expressions, JSON, branching, loops, and reusable sub-workflows.

03 Integrate

Connect REST APIs, webhooks, apps, and PostgreSQL with suitable authentication.

04 Apply AI

Build structured LLM workflows and a grounded knowledge assistant.

05 Operate

Test edge cases, handle errors, protect secrets, and monitor executions.

06 Deliver

Deploy a small solution, document it, and explain its business value.

How each 3-hour session works

A consistent rhythm balances explanation, guided practice, and independent problem-solving.

00–30 MIN	**RECAP & CONCEPT**	Connect the topic to last session; explain the mental model.
30–75 MIN	**LIVE BUILD**	Instructor demonstrates a small workflow and explains decisions.
75–135 MIN	**HANDS-ON LAB**	Learners build, modify, and debug a guided task.
135–165 MIN	**CHALLENGE**	Complete an extension or handle a realistic edge case.
165–180 MIN	**REVIEW**	Test results, discuss failures, and set the next task.

Every session ends with a tangible output, a debugging lesson, or a measurable checkpoint.

Six stages. One end-to-end skill set

Each stage contains 6 sessions × 3 hours = 18 guided hours.

01 FOUNDATIONS & N8N BASICS

Sessions 01–06 — **18 hours**

02 WORKFLOW LOGIC & DATA

Sessions 07–12 — **18 hours**

03 APIs, WEBHOOKS & DATABASES

Sessions 13–18 — **18 hours**

04 LLMS & AI WORKFLOWS

Sessions 19–24 — **18 hours**

05 RAG, AGENTS & RELIABILITY

Sessions 25–30 — **18 hours**

06 DEPLOYMENT & CAPSTONE

Sessions 31–36 — **18 hours**

Automation mindset & n8n orientation

Three 3-hour sessions. Each session includes topic coverage and a practical output.

SESSION 01

Automation thinking & process mapping

Automation vs AI automation; identify repetitive tasks; triggers, actions, inputs, outputs; map a business process.

PRACTICAL — Draw a process map and choose an automation candidate.

SESSION 02

n8n interface & first workflow

Cloud vs self-hosted overview; canvas, nodes, connections, executions; manual trigger; activate vs test.

PRACTICAL — Build a first workflow and inspect its execution.

SESSION 03

Triggers, schedules & notifications

Manual and Schedule triggers; basic app actions; workflow naming; execution history; safe test messages.

PRACTICAL — Create a scheduled reminder or notification workflow.

MODULE 01 · FOUNDATIONS & N8N BASICS

Workflow anatomy & working with data

Three 3-hour sessions. Each session includes topic coverage and a practical output.

SESSION 04

Items, JSON & data flow

JSON objects and arrays; items; node input/output; field paths; inspect sample payloads.

PRACTICAL — Trace a record through connected nodes.

SESSION 05

Expressions & dynamic values

Expression editor; referencing previous nodes; string/date formatting; fallback values; common errors.

PRACTICAL — Create personalized messages from dynamic fields.

SESSION 06

Build Lab 01: office automation

Combine trigger, data mapping, notification; workflow naming; test cases; review and feedback.

PRACTICAL — Mini-project: form or schedule → transform data → notify.

MODULE 01 · FOUNDATIONS & N8N BASICS

Conditions, routing & data transformation

Three 3-hour sessions. Each session includes topic coverage and a practical output.

SESSION 07

IF, Switch & filters

Conditional branches; comparisons; null/missing values; routing by category or priority.

PRACTICAL — Route incoming requests to different paths.

SESSION 08

Merge, loops & batch processing

Merge strategies; looping through items; batching; aggregation; avoid unnecessary loops.

PRACTICAL — Process a list of records and combine results.

SESSION 09

Code node essentials

When to use Code; JavaScript basics; arrays and objects; map/filter patterns; readable logic.

PRACTICAL — Transform a messy dataset in a Code node.

Reusable workflows & dependable execution

Three 3-hour sessions. Each session includes topic coverage and a practical output.

SESSION 10

Dates, files & data cleanup

Date/time handling; text normalization; CSV/spreadsheet data; binary data basics.

PRACTICAL — Normalize contact data and prepare a clean export.

SESSION 11

Sub-workflows & reusable components

Call a sub-workflow; define inputs/outputs; modular design; avoid duplicated logic.

PRACTICAL — Extract a repeated task into a reusable workflow.

SESSION 12

Errors, retries & debugging

Read node errors; retry transient failures; Error Trigger basics; alerting; safe failure paths.

PRACTICAL — Build an error-notification path and test a deliberate failure.

MODULE 02 · WORKFLOW LOGIC & DATA

HTTP, REST APIs & authentication

Three 3-hour sessions. Each session includes topic coverage and a practical output.

SESSION 13

HTTP and REST API fundamentals

Requests/responses; GET/POST/PATCH/DELETE; headers, query parameters, body; status codes.

PRACTICAL — Call a public API and extract useful fields.

SESSION 14

HTTP Request node in practice

Authentication options; pagination; response parsing; rate limits; timeout and retry strategy.

PRACTICAL — Collect API results into a clean dataset.

SESSION 15

API authentication & credentials

API keys, bearer tokens, OAuth overview; credential storage; scopes and least privilege.

PRACTICAL — Connect a service using secure credentials.

Webhooks, apps & persistence

Three 3-hour sessions. Each session includes topic coverage and a practical output.

SESSION 16

Webhooks & event-driven workflows

Test vs production URLs; payload validation; event-driven design; public endpoints and tunnel concepts.

PRACTICAL — Receive a webhook and validate its data.

SESSION 17

App integrations & cross-system sync

Email, spreadsheets, forms, chat, calendar or CRM integrations; map fields between systems.

PRACTICAL — Synchronize records between two services.

SESSION 18

PostgreSQL & data persistence

Tables and keys; basic SQL CRUD; parameterized queries; store workflow status; deduplication.

PRACTICAL — Save webhook events to PostgreSQL and prevent duplicates.

MODULE 03 · APIS, WEBHOOKS & DATABASES

LLM foundations & structured AI outputs

Three 3-hour sessions. Each session includes topic coverage and a practical output.

SESSION 19

LLMs for automation: strengths & limits

Tokens and context; model selection; cost and latency; hallucinations; use rules/code when better.

PRACTICAL — Choose where AI helps in a sample business workflow.

SESSION 20

Prompt design & structured outputs

System/user prompts; examples; constraints; JSON schema concepts; parse and validate output.

PRACTICAL — Classify messages into validated JSON.

SESSION 21

AI-powered text & document workflows

Summarization, extraction, classification, sentiment; document inputs; confidence and fallback logic.

PRACTICAL — Build an AI email or document triage workflow.

MODULE 04 · LLMS & AI WORKFLOWS

AI nodes, tools & human oversight

Three 3-hour sessions. Each session includes topic coverage and a practical output.

SESSION 22

AI nodes, chains & model connections

Connect a supported model; prompt templates; pass context; inspect model outputs and errors.

PRACTICAL — Create a prompt-driven workflow with structured results.

SESSION 23

AI agents & tool calling fundamentals

Agent vs chain; tools and descriptions; constrained tool access; stopping conditions; practical limits.

PRACTICAL — Build a small agent that uses approved tools.

SESSION 24

Human approval & AI workflow project

Approval gates; draft-before-send; escalation; prompt injection awareness; evaluate outputs.

PRACTICAL — AI assistant drafts an action for human approval.

MODULE 04 · LLMS & AI WORKFLOWS

Knowledge bases & RAG

Three 3-hour sessions. Each session includes topic coverage and a practical output.

SESSION 25

Document ingestion & retrieval design

PDF/text ingestion; extraction; chunking; metadata; source documents and access considerations.

PRACTICAL — Prepare a small searchable knowledge collection.

SESSION 26

Embeddings & vector search concepts

What embeddings represent; similarity search; vector database options; metadata filters.

PRACTICAL — Create embeddings and retrieve relevant chunks.

SESSION 27

Build a grounded RAG assistant

Retrieval → context → answer; source references; unknown-answer fallback; evaluation examples.

PRACTICAL — Answer questions from a small policy or FAQ knowledge base.

MODULE 05 · RAG, AGENTS & RELIABILITY

Agent workflows, safety & testing

Three 3-hour sessions. Each session includes topic coverage and a practical output.

SESSION 28

Agent workflow design patterns

Tool selection; structured tool inputs; persistent data vs conversational memory; limits and timeouts.

PRACTICAL — Create a bounded agent that retrieves records and recommends next steps.

SESSION 29

Security, privacy & safe automation

Credential protection; least privilege; sensitive data; untrusted inputs; prompt injection; approval for consequential actions.

PRACTICAL — Threat-model an AI workflow and add safety controls.

SESSION 30

Testing, monitoring & recovery

Happy-path and edge-case tests; retries vs duplicate side effects; logs; alerts; runbook basics.

PRACTICAL — Test failures and produce a monitoring/recovery plan.

MODULE 05 · RAG, AGENTS & RELIABILITY

Deployment & operating a small production system

Three 3-hour sessions. Each session includes topic coverage and a practical output.

SESSION 31

Deployment choices & Docker basics

n8n Cloud vs self-hosting; environment variables; Docker and Compose concepts; persistent volumes.

PRACTICAL — Run a simple containerized service or map a deployment plan.

SESSION 32

Domains, HTTPS & secure configuration

Public webhook URLs; DNS and reverse proxy concepts; HTTPS; firewall basics; secure access.

PRACTICAL — Prepare a deployment checklist for a small production workflow.

SESSION 33

Backups, upgrades & maintenance

PostgreSQL backups; restore concepts; versioning; update planning; execution retention; operating costs.

PRACTICAL — Write a backup, update, and recovery checklist.

MODULE 06 · DEPLOYMENT & CAPSTONE

Capstone integration, review & presentation

Three 3-hour sessions. Each session includes topic coverage and a practical output.

SESSION 34

Capstone build sprint

Combine triggers, APIs, data persistence, AI where appropriate, and notifications; define acceptance criteria.

PRACTICAL — Working end-to-end capstone workflow.

SESSION 35

Capstone testing & delivery hardening

Test invalid data, duplicates, API failures, AI uncertainty, approvals; documentation; demo preparation.

PRACTICAL — Release candidate with tests, README, and workflow export.

SESSION 36

Project showcase & professional handover

Demonstration; architecture explanation; value and limitations; peer review; maintenance and next steps.

PRACTICAL — Present capstone, test evidence, and handover documentation.

MODULE 06 · DEPLOYMENT & CAPSTONE

Capstone: an AI-enabled business workflow

Learners select a realistic business problem and build a complete, testable automation around it.

01 CAPTURE

Form, webhook, email, or scheduled input

02 VALIDATE

Check fields, normalize data, detect duplicates

03 ENRICH

Call an API or retrieve business context

04 DECIDE

Use rules first; AI for language tasks when useful

05 APPROVE

Human approval for sensitive or external actions

06 EXECUTE & LOG

Update systems, notify, and record outcome

REQUIRED CAPSTONE EVIDENCE

Workflow export, architecture diagram, test cases, failure handling, README, and a short demo.

SUCCESS CRITERIA

Solves a real problem; handles common failures; uses credentials safely; explains impact and limits.

Assessment built around demonstrated ability

Suggested grading model for an institution-led delivery; weights can be adapted to local policy.

WEEKLY LABS & CHECKPOINTS	25%	Small workflows, debugging tasks, and short reflections.
INTEGRATION PROJECT	20%	API/webhook/database workflow with validation and error handling.
AI WORKFLOW PROJECT	20%	Structured AI output or RAG assistant with evaluation and safe fallback.
CAPSTONE & DEMO	30%	End-to-end solution, testing, documentation, and presentation.
PARTICIPATION & REVIEW	5%	Peer feedback, technical explanation, and responsible practice.

Recommended minimum: submit major projects and demonstrate that the workflow works—not only that it exists.

Tools learners will work with

Use free tiers and local development where possible; paid APIs or hosting may be optional.

CORE PLATFORM

n8n; official documentation; workflow templates.

DATA & INTEGRATIONS

JSON; HTTP/REST; webhooks; spreadsheets; email/chat apps; PostgreSQL.

AI COMPONENTS

An LLM API or compatible model; structured outputs; embeddings; a vector store for RAG.

ENGINEERING

Git/GitHub; basic JavaScript; optional Python; Docker basics; logs and test data.

DEPLOYMENT

n8n Cloud or self-hosted instance; domain/HTTPS for relevant deployment exercises.

COST & ACCESS

API accounts may require billing; use test data and never share secrets.

From workflow builder to automation professional

A repeatable delivery process for freelance, internal, or product work.

DISCOVER

Understand the current process, pain points, users, and success metric.

DESIGN

Choose triggers, data sources, permissions, failure paths, and acceptance criteria.

BUILD

Implement in small steps; use reusable components and versioned exports.

VERIFY

Test valid, invalid, duplicate, timeout, and AI-uncertain cases.

HAND OVER

Provide README, architecture, ownership, runbook, and demo.

IMPROVE

Monitor results, estimate value, and refine based on real usage.

A professional deliverable is tested, explainable, secure, documented, and maintainable—not merely a workflow that ran once.

Professional scope—without unnecessary extreme complexity

Target: practical independence and small-to-medium business delivery, not hyperscale infrastructure engineering.

INCLUDED IN THIS COURSE

- Core n8n workflow design and debugging
- REST APIs, webhooks, credentials, PostgreSQL
- LLM prompts and structured outputs
- Introductory agents with constrained tools
- RAG fundamentals and grounded answers
- Human approval, testing, error handling
- Docker and small-scale deployment basics
- Documentation, handover, and business value

NOT A CORE REQUIREMENT

- Multi-agent orchestration architectures
- Large-scale queue/worker tuning
- Kubernetes or complex cloud infrastructure
- Training or fine-tuning foundation models
- Building a vector database from scratch
- Fully autonomous high-risk actions
- Deep distributed-systems engineering

Advanced topics can be explored later as optional extensions after learners can build and maintain reliable systems.

What learners leave with

A portfolio of evidence showing technical ability and practical judgement.

WORKFLOW PORTFOLIO

3–5 exported workflows with clear names, sample inputs, and documented outputs.

INTEGRATION BUILD

A workflow connecting an API/webhook to persistent storage with validation.

AI BUILD

An AI-assisted workflow or grounded RAG assistant with safe fallback behavior.

CAPSTONE

A working end-to-end automation for a realistic business process.

TECHNICAL DOCUMENTATION

README, architecture diagram, test cases, limitations, and maintenance notes.

DEMO & CASE STUDY

Short demonstration explaining problem, solution, results, and next improvements.

A clear 12-week progression

Each two-week block builds toward the capstone, with practical work from week one.

W01–02

FOUNDATIONS

6 sessions · 18h

W03–04

WORKFLOW LOGIC

6 sessions · 18h

W05–06

INTEGRATIONS

6 sessions · 18h

W07–08

AI WORKFLOWS

6 sessions · 18h

W09–10

RAG & RELIABILITY

6 sessions · 18h

W11–12

DEPLOY & CAPSTONE

6 sessions · 18h

12 weeks × 3 sessions/week × 3 hours/session = 108 guided hours.

Build useful systems. Deliver real outcomes.

A zero-to-professional learning journey in workflow automation, AI integration, and responsible delivery.

36 SESSIONS • 108 HOURS • 12 WEEKS

Thank you



FROM ZERO TO REAL-WORLD AUTOMATION